

A 30-year-old patient with myotonic dystrophy is scheduled for surgery for acute appendicitis. Outline your anaesthetic considerations.
(Modified from FANZCA 2016A)

Examiner Report

Borderline standard: To pass the question the candidate had to demonstrate that they understood the condition and the risk anaesthesia and surgery poses for the patient. Anaesthesia management needed to reflect this understanding and include at least respiratory issues.

Comment: Some candidates discussed congenital muscle disease in general, without referring to myotonic dystrophy. While some marks were awarded those who could discuss myotonic dystrophy specifically performed better.

Sample Answer

This is a young patient with myotonic dystrophy presenting for emergency abdominal surgery. Myotonic dystrophy is an autosomal dominant myotonia disorder which poses significant anaesthetic challenges, in particular cardiorespiratory management. Anaesthetic considerations include:

Patient Factors

- Airway
 - Aspiration risk due to weakness of the pharyngeal muscles from progressive bulbar palsy.
 - Evaluate for potential difficult airway (rescue bag-mask ventilation may be challenging)
 - Review previous anaesthesia charts (if available) for Cormack & Lehane grading
- Respiratory
 - May be associated with restrictive lung disease and OSA.
 - HD for post-op monitoring, KIV ventilatory support post-op in ICU.
 - Incentive spirometry if time permits.
 - Assessment of effort tolerance, ability to cough, oxygen saturations on room air, and recent spirometry results (if available) will be useful in determining disposition and need for ventilatory support post-op.
- Cardiovascular
 - May have cardiomyopathy, heart failure, arrhythmias, mitral valve prolapse (20%)
 - Review cardiac history (palpitation, syncope, angina, effort tolerance), 12-lead ECG, 2DE (if available). Cardiology consult for optimisation if time permits.
 - Check for pacemaker/ ICD in-situ, and take peri-op precautions as necessary (interrogation/ re-programming/ magnet placement/ defib pads).
 - Defibrillator pads may be placed if high risk for malignant arrhythmias.
 - Arrhythmias should be treated aggressively intra-op.
 - Electrolyte abnormalities, in particular K and Mg, should be corrected pre-op.
 - KIV 5-lead ECG, IA line may be required if cardiac concerns are present.
- Gastrointestinal
 - Delayed gastric emptying further contributes to increased aspiration risk
 - Use of PPI and antacid (sodium citrate) pre-operatively may minimise aspiration pneumonitis
- Neurology
 - Behavioural and cognitive problems may be present, making counselling more difficult and patient cooperation (for awake procedures) more challenging.
 - Narcolepsy-like central hypersomnia/ central sleep apnoea increases sensitivity to opioids.
- Endocrinology
 - Associated with diabetes mellitus, hypothyroidism, and adrenal insufficiency.

- Endocrine referral for peri-operative management and optimization.
 - May require stress dose steroids, with omeprazole cover.
- Myotonia
 - Myotonic episodes may be triggered by a variety of causes during anaesthesia
 - Offending drugs include succinylcholine, anti-cholinesterases, and opioids.
 - Other triggers are hypothermia, shivering or surgical stimulation.
 - Care should be taken to avoid these triggers of myotonic episodes intra-op.
 - Anti-myotonic medications should be continued peri-operatively
- Musculoskeletal
 - Care with intra-op positioning in view of contractures/myotonia to prevent pressure sores and neuropraxia.

Anaesthesia factors

- Multi-disciplinary team approach in managing this patient peri-operatively is essential to ensure safe and successful anaesthesia.
- Avoid sedative premedication such as benzodiazepines. Titrate depth of anaesthesia with BIS target 40-60.
- Care with laryngoscopy and jaw manipulation due to risk of temporomandibular joint dislocation
- Myotonic dystrophy not known to be associated with malignant hyperthermia.
 - Rocuronium preferred over Suxamethonium (known trigger for myotonia, hyperkalaemia)
 - Volatile anaesthetics may be used for maintenance.
- Discuss with surgeon on use of RA such as TAP block or local infiltrative techniques as part of opioid-sparing analgesia.
- Temperature monitoring, with use of passive and active warming devices such as Bair Hugger to maintain normothermia.
- Strict extubation criteria necessary due to risks of respiratory failure and aspiration
 - Pharyngeal weakness with poor airway protection
 - Altered gastric motility
 - Cardiopulmonary pathology
- TOF monitor with TOF ratio >0.9 may be used to guide muscle recovery, but care should be taken during use (may trigger myotonia)
- Use of neostigmine for reversal may cause myotonia
 - Sugammadex may be used for reversal (if Rocuronium was used for maintenance)
- Opioid-sparing analgesia regime should be continued for post-op pain. If opioid has to be initiated/continued post-operatively, then patient should be monitored in HD/ ICU.
- Standby phenytoin/ procainamide for myotonic crisis.

Surgical factors

- Limited time for optimization of patient's condition due to emergent nature of surgery
 - Hydration may be maintained with IV fluids during NBM period
- Discussion with surgeons on surgical approach (laparoscopic vs open)
 - RA (Spinal/ Epidural) may be used for open approach
 - GA would be required for laparoscopic approach
- Acute abdomen (aspiration risk) necessitates RSI.